



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

School of Electrical Engineering and Computer Science

CS-250: Data Structure and Algorithms

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Class		BSDS-3A

Lab 4: Doubly & Circular Linked List

Date: 29th September 2026

Time: 10:00 am – 12:50 pm

Instructor: Dr. Fahad Ahmed Satti

Lab Engineer: Hassan Mohtashim



GitHub Link:

<https://github.com/tsandhubsds25seecs-code/LAB5-TAYYABALISANDHU-540469>

Task 1: Browser Tab Manager

Implement a browser tab manager using a Circular Doubly Linked List. Each node represents an open browser tab. A current pointer identifies the active tab. Each node should store:

- Tab ID (integer)
- Website Title (string)
- URL (string)

Your program should support the following operations:

1. **Open New Tab** – Insert a new tab after the current tab. If the list is empty, the new tab becomes the first and current tab.
2. **Close Current Tab** – Remove the current tab. The next tab becomes current. If it was the only tab, the list becomes empty.
3. **Move Next** – Move current to current -> next.
4. **Move Previous** – Move current to current -> prev.
5. **Display Current Tab** – Show the ID, title, and URL of the current tab.
6. **Display All Tabs Forward** – Starting from current, display each tab exactly once.
7. **Display All Tabs Backward** – Starting from current, display each tab exactly once in reverse direction.
8. **Search Tab** – Search for a tab by its ID and display its details.

Implementation constraints:

- Do not use arrays, vectors, or other built-in containers to store the tabs.
- Forward and backward display must terminate by detecting a return to the starting node, not by checking for NULL.
- After every operation, the CDLL must remain circular and doubly connected.



Problems Output Debug Console **Terminal** Ports 1

```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task1
```

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 1

Enter tab title: google

Enter tab URL: www.google.com

Tab opened: google (www.google.com)

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 1

Enter tab title: GitHub

Enter tab URL: www.github.com

Tab opened: GitHub (www.github.com)



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task1
```

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 2

Tab closed: GitHub (www.github.com)

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 2

Tab closed: google (www.google.com)



tsandhu_@TAYYAB-ALI-SANDHU:~\$./task1

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 2

No tabs to close.

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 1

Enter tab title: Youtube

Enter tab URL: www.youtube.com

Tab opened: Youtube (www.youtube.com)

Browser Tab Manager

1. Open Tab



Problems Output Debug Console Terminal Ports 1

```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task1
```

```
Tab opened: Youtube (www.youtube.com)
```

```
Browser Tab Manager
```

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

```
Enter your choice: 1
```

```
Enter tab title: lms
```

```
Enter tab URL: www.lms.nust.edu.pk
```

```
Tab opened: lms (www.lms.nust.edu.pk)
```

```
Browser Tab Manager
```

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

```
Enter your choice: 3
```

```
Moved to tab: Youtube (www.youtube.com)
```

```
Browser Tab Manager
```



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task1
```

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 4

Moved to tab: lmS (www.lms.nust.edu.pk)

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 5

Current tab: lmS (www.lms.nust.edu.pk)

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab



tsandhu_@TAYYAB-ALI-SANDHU:~\$./task1

6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 6

All tabs (forward):

3. Youtube (www.youtube.com)
4. lms (www.lms.nust.edu.pk)

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit

Enter your choice: 7

All tabs (backward):

4. lms (www.lms.nust.edu.pk)
3. Youtube (www.youtube.com)

Browser Tab Manager

1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward



```
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit
Enter your choice: 8
Enter tab ID to search: 3
Tab found: Youtube (www.youtube.com)
```

```
Browser Tab Manager
```

```
1. Open Tab
2. Close Current Tab
3. Move to Next Tab
4. Move to Previous Tab
5. Display Current Tab
6. Display All Tabs Forward
7. Display All Tabs Backward
8. Search Tab by ID
9. Exit
Enter your choice: 9
Exiting...
```

```
❖ tsandhu_@TAYYAB-ALI-SANDHU:~$
```



Task 2: Circular Photo Album

Implement a **photo album management system using a Circular Doubly Linked List**. Each photo is represented by a node containing information about the photo. The album should allow the user to navigate continuously in both forward and backward directions. Each node should store:

- Photo ID (integer)
- Photo Name (string)
- Date Taken (string)
- Location (string)

The program should take as input:

- Number of photos
- Details of each photo

Your program should support the following operations:

1. **Add Photo** – Insert a new photo at the end of the circular doubly linked list.
2. **Insert Photo After Current** – Insert a new photo immediately after the currently selected photo.
3. **Remove Photo** – Remove a photo using its Photo ID.
4. **Remove Current Photo** – Remove the currently selected photo. After deletion, the next photo should become the current photo.
5. **Move Next** – Move the current photo to the next photo using the next pointer.
6. **Move Previous** – Move the current photo to the previous photo using the prev pointer.
7. **Display Album Forward** – Display all photos exactly once, starting from the current photo and moving through the next pointers.
8. **Display Album Backward** – Display all photos exactly once, starting from the current photo and moving through the prev pointers.
9. **Search Photo** – Search for a photo using its Photo ID and display its details.
10. **Count Photos** – Count and display the total number of photos currently present in the album.

Implementation constraints:

1. The album must be implemented only using a Circular Doubly Linked List.
2. Each node must contain both next and prev pointers.
3. The last node's next pointer must point to the first node.
4. The first node's prev pointer must point to the last node.
5. Forward traversal must use the next pointer.
6. Backward traversal must use the prev pointer.



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7. Do not use arrays, vectors, or other built-in containers to store the photos.
8. Traversal must terminate when the traversal reaches the starting node again; do not use NULL as the termination condition for a non-empty list.
9. The program must correctly handle an empty album and an album containing only one photo.
10. After every insertion and deletion, the circular and doubly linked relationships must remain valid.
11. Dynamically allocated memory for deleted nodes must be properly released. After every operation, the CDLL must remain circular and doubly connected.

Outputs:

```
tsandhu_@TAYYAB-ALI-SANDHU:~$ g++ -std=c++17 -Wall task2.cpp -o task2
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task2
Enter number of photos: 5

Photo 1:
Enter photo name: Ahmad
Enter date taken: 10/10/2026
Enter location: Islamabad
Photo added: Ahmad (Islamabad)

Photo 2:
Enter photo name: Nust
Enter date taken: 11/10/2026
Enter location: Islamabad h12
Photo added: Nust (Islamabad h12)

Photo 3:
Enter photo name: laptop
Enter date taken: 112/10/2026
Enter location: murree
Photo added: laptop (murree)

Photo 4:
Enter photo name: Hiran minar
Enter date taken: 10/09/2026
Enter location: Sheikhpura
Photo added: Hiran minar (Sheikhpura)

Photo 5:
Enter photo name: fort
Enter date taken: 12/12/2025
Enter location: Lahore
Photo added: fort (Lahore)

Photo Album Manager
1. Add Photo
```



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task2
```

2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo
6. Move to Previous Photo
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit

Enter your choice: 3

Enter photo ID to remove: 3

Photo removed: laptop

Photo Album Manager

1. Add Photo
2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo
6. Move to Previous Photo
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit

Enter your choice: 5

Moved to photo: Nust (Islamabad h12)

Photo Album Manager

1. Add Photo
2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task2
```

```
6. Move to Previous Photo
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit
Enter your choice: 6
Moved to photo: Ahmad (Islamabad)
```

```
Photo Album Manager
```

```
1. Add Photo
2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo
6. Move to Previous Photo
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit
Enter your choice: 7
Album (forward):
1. Ahmad | 10/10/2026 | Islamabad
2. Nust | 11/10/2026 | Islamabad h12
4. Hiran minar | 10/09/2026 | Sheikhupura
5. fort | 12/12/2025 | Lahore
```

```
Photo Album Manager
```

```
1. Add Photo
2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo
6. Move to Previous Photo
```



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task2
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit
Enter your choice: 8
Album (backward):
1. Ahmad | 10/10/2026 | Islamabad
5. fort | 12/12/2025 | Lahore
4. Hiran minar | 10/09/2026 | Sheikhupura
2. Nust | 11/10/2026 | Islamabad h12

Photo Album Manager
1. Add Photo
2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo
6. Move to Previous Photo
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit
Enter your choice: 9
Enter photo ID to search: 10
Photo with ID 10 not found.

Photo Album Manager
1. Add Photo
2. Insert Photo After Current
3. Remove Photo by ID
4. Remove Current Photo
5. Move to Next Photo
6. Move to Previous Photo
```



```
7. Display Album Forward
8. Display Album Backward
9. Search Photo by ID
10. Count Photos
11. Exit
Enter your choice: 11
Exiting...
tsandhu_@TAYYAB-ALI-SANDHU:~$
```

Task 3: Train Coach Navigation System

Implement a **train coach management system** using a Circular Doubly Linked List. The train is circular, so after the last coach comes the first coach. Because the structure is doubly linked, the train can also be traversed in the reverse direction.

Each node should store:

- Coach Number (integer)
- Coach Type (string)
- Passenger Capacity (integer)
- Current Passengers (integer)

The program should take as input:

- Number of coaches
- Details of each coach

Your program should support the following operations:

1. **Add Coach** – Add a coach at the end of the train.
2. **Insert Coach** – Insert a coach after a specified coach number.
3. **Remove Coach** – Remove a coach-by-coach number.
4. **Move Forward** – Move the current coach to the next coach.
5. **Move Backward** – Move the current coach to the previous coach.
6. **Display Train Clockwise** – Display all coaches exactly once.
7. **Display Train Anti-clockwise** – Display all coaches exactly once in reverse direction.
8. **Search Coach** – Search by coach number.
9. **Find Maximum Available Capacity** – Find and display the coach with the largest number of empty seats.
10. **Display Current Coach** – Display the coach currently selected.



11. **Reverse Train Direction** – Reverse the direction of the complete CDLL by manipulating pointers, not by copying node data to another structure.

Implementation constraints:

1. The train must be implemented only using a Circular Doubly Linked List.
2. Each node must contain both next and prev pointers.
3. The last coach's next pointer must point to the first coach.
4. The first coach's prev pointer must point to the last coach.
5. Forward traversal must use the next pointer.
6. Backward traversal must use the prev pointer.
7. Do not use arrays, vectors, or other built-in containers to store the coaches.
8. Traversal must terminate when the traversal reaches the starting coach again; do not use NULL as the termination condition for a non-empty list.
9. The program must correctly handle an empty train and a train containing only one coach.
10. After every insertion and deletion, the circular and doubly linked relationships must remain valid.
11. When removing the current coach, the next valid coach should become the current coach.
12. The Reverse Train Direction operation must modify the links of the existing nodes rather than copying the coach data into another data structure.
13. After reversing the train, both forward and backward traversal must continue to work correctly.
14. Dynamically allocated memory for deleted coaches must be properly released.



```
Problems Output Debug Console Terminal Ports 3
● tsandhu_@TAYYAB-ALI-SANDHU:~$ g++ -std=c++17 -Wall task3.cpp -o t
○ tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task3
Enter number of coaches: 4

Coach 1:
Enter coach number: 009
Enter coach type: student
Enter capacity: 50
Enter current passengers: 45
Coach added: 9 (student)

Coach 2:
Enter coach number: 9090
Enter coach type: Faculty
Enter capacity: 30
Enter current passengers: 29
Coach added: 9090 (Faculty)

Coach 3:
Enter coach number: 8080
Enter coach type: passengers
Enter capacity: 80
Enter current passengers: 78
Coach added: 8080 (passengers)

Coach 4:
Enter coach number: 1010
Enter coach type: army
Enter capacity: 30
Enter current passengers: 30
Coach added: 1010 (army)
```



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```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task3
```

```
Coach added: 1010 (army)
```

```
Train Coach Navigation System
```

1. Add Coach
2. Insert Coach After
3. Remove Coach
4. Move Forward
5. Move Backward
6. Display Train Clockwise
7. Display Train Anti-clockwise
8. Search Coach
9. Find Max Available Capacity
10. Display Current Coach
11. Reverse Train Direction
12. Exit

```
Enter your choice: 3
```

```
Enter coach number to remove: 9
```

```
Coach removed: 9
```

```
Train Coach Navigation System
```

1. Add Coach
2. Insert Coach After
3. Remove Coach
4. Move Forward
5. Move Backward
6. Display Train Clockwise
7. Display Train Anti-clockwise
8. Search Coach
9. Find Max Available Capacity
10. Display Current Coach
11. Reverse Train Direction
12. Exit

```
Enter your choice: 6
```

```
Train (clockwise):
```

```
Coach 9090 | Faculty | Capacity: 30 | Passengers: 29 | Empty: 1
```



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task3
```

```
12. Exit
```

```
Enter your choice: 6
```

```
Train (clockwise):
```

```
Coach 9090 | Faculty | Capacity: 30 | Passengers: 29 | Empty: 1
```

```
Coach 8080 | passengers | Capacity: 80 | Passengers: 78 | Empty:
```

```
Coach 1010 | army | Capacity: 30 | Passengers: 30 | Empty: 0
```

```
Train Coach Navigation System
```

```
1. Add Coach
```

```
2. Insert Coach After
```

```
3. Remove Coach
```

```
4. Move Forward
```

```
5. Move Backward
```

```
6. Display Train Clockwise
```

```
7. Display Train Anti-clockwise
```

```
8. Search Coach
```

```
9. Find Max Available Capacity
```

```
10. Display Current Coach
```

```
11. Reverse Train Direction
```

```
12. Exit
```

```
Enter your choice: 7
```

```
Train (anti-clockwise):
```

```
Coach 1010 | army | Capacity: 30 | Passengers: 30 | Empty: 0
```

```
Coach 8080 | passengers | Capacity: 80 | Passengers: 78 | Empty:
```

```
Coach 9090 | Faculty | Capacity: 30 | Passengers: 29 | Empty: 1
```

```
Train Coach Navigation System
```

```
1. Add Coach
```

```
2. Insert Coach After
```

```
3. Remove Coach
```

```
4. Move Forward
```

```
5. Move Backward
```

```
6. Display Train Clockwise
```

```
7. Display Train Anti-clockwise
```

```
8. Search Coach
```



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task3
```

4. Move Forward
5. Move Backward
6. Display Train Clockwise
7. Display Train Anti-clockwise
8. Search Coach
9. Find Max Available Capacity
10. Display Current Coach
11. Reverse Train Direction
12. Exit

Enter your choice: 5

Moved backward to coach: 9090

Train Coach Navigation System

1. Add Coach
2. Insert Coach After
3. Remove Coach
4. Move Forward
5. Move Backward
6. Display Train Clockwise
7. Display Train Anti-clockwise
8. Search Coach
9. Find Max Available Capacity
10. Display Current Coach
11. Reverse Train Direction
12. Exit

Enter your choice: 11

Train direction reversed.

Train Coach Navigation System

1. Add Coach
2. Insert Coach After
3. Remove Coach
4. Move Forward
5. Move Backward
6. Display Train Clockwise



```
tsandhu_@TAYYAB-ALI-SANDHU:~$ ./task3
```

```
8. Search Coach
```

```
9. Find Max Available Capacity
```

```
10. Display Current Coach
```

```
11. Reverse Train Direction
```

```
12. Exit
```

```
Enter your choice: 9
```

```
Coach with max empty seats: 8080 (passengers) | Empty seats: 8080
```

```
Train Coach Navigation System
```

```
1. Add Coach
```

```
2. Insert Coach After
```

```
3. Remove Coach
```

```
4. Move Forward
```

```
5. Move Backward
```

```
6. Display Train Clockwise
```

```
7. Display Train Anti-clockwise
```

```
8. Search Coach
```

```
9. Find Max Available Capacity
```

```
10. Display Current Coach
```

```
11. Reverse Train Direction
```

```
12. Exit
```

```
Enter your choice: 11
```

```
Train direction reversed.
```

```
Train Coach Navigation System
```

```
1. Add Coach
```

```
2. Insert Coach After
```

```
3. Remove Coach
```

```
4. Move Forward
```

```
5. Move Backward
```

```
6. Display Train Clockwise
```

```
7. Display Train Anti-clockwise
```

```
8. Search Coach
```

```
9. Find Max Available Capacity
```

```
10. Display Current Coach
```



```
Enter your choice: 11
Train direction reversed.
```

```
Train Coach Navigation System
```

1. Add Coach
2. Insert Coach After
3. Remove Coach
4. Move Forward
5. Move Backward
6. Display Train Clockwise
7. Display Train Anti-clockwise
8. Search Coach
9. Find Max Available Capacity
10. Display Current Coach
11. Reverse Train Direction
12. Exit

```
Enter your choice: 12
Exiting...
```

```
tsandhu_@TAYYAB-ALI-SANDHU:~$
```